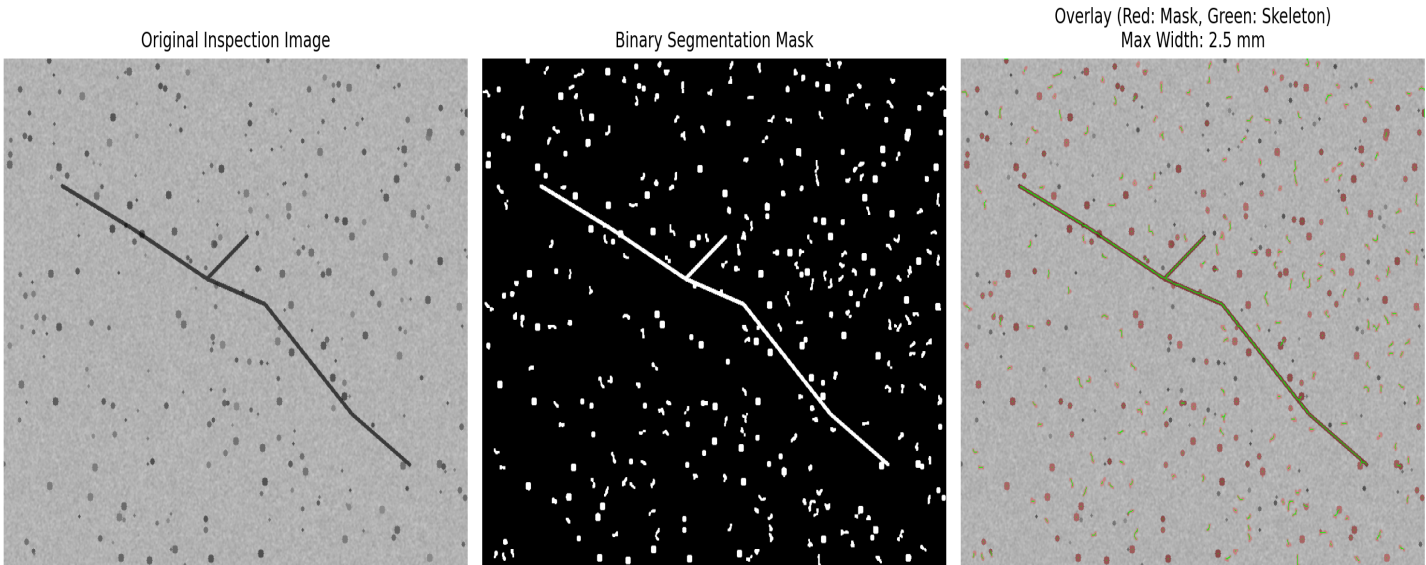


AI-BASED STRUCTURAL CRACK INSPECTION REPORT

Automated Computer Vision Quantitative Assessment & Risk Analysis

Project:	Civil Infrastructure Health Monitoring	Date/Time:	2026-08-25 20:26
Structure Element:	RC-Girder-G12	Exposure Class:	humid_air
Inspector / Firm:	Automated AI Vision System	Scale Ratio:	0.25 mm/px

1. Visual Diagnostic Overlay



2. Geometric Quantification (OpenCV Engine)

Metric Parameter	Value	Engineering Significance
Total Crack Length (\$L_c\$)	607.5 mm	Centerline medial axis tracking
Maximum Crack Width (\$w_{\max}\$)	2.5 mm	Critical ACI 224R / Eurocode 2 metric
Average Crack Width (\$w_{\text{avg}}\$)	1.21 mm	Permeability profile estimation
Total Damaged Area	1366.19 mm²	Surface spalling and damage extent

3. Structural Risk Assessment & Action Plan

Severity Classification:	Critical / Structural Concern
Risk Code:	HIGH-3
Governing Standard:	Eurocode 2 / ACI 318 / ACI 224.1R
Serviceability Impact:	High risk to structural durability, integrity, and shear/flexural serviceability.
Recommended Action:	Immediate non-destructive testing (NDT/Ultrasonic), core testing, and structural engineering appraisal for potential strengthening (FRP/jacketing).